Battery Drain by Location Accuracy Report

By Christopher Paquin

This report will be a summary of my findings related to the battery drain of cell phones from using varying degrees of precision in GPS. GPS is mandatory for the use of the IBDC app so picking the right precision will be an important balance of accuracy and attrition cost. My recommendation is at the bottom of the report.

Assumptions:

- The GPS will be active for the entire duration of any bike ride. This is a safe assumption as the app won't provide its intended benefit without GPS.
- All precision options are available to us. This is a safe assumption as these are simply a request for a permission level by the app to the user which they can approve or deny. It's reasonable to assume they know the app won't work effectively without GPS.
- A bike ride may last up to 5 hours. A bike ride is very likely to be longer than a typical run, but I was unable to find any distribution graphs for road cycling times so this is based on what I saw as the upper common limit from the Facebook posts of an avid cycling friend.

Accuracy Levels

According to Android & IOS documentation there are x android and y IOS levels of locations data we can request in code with varying accuracy tolerances and battery costs. An important note is that just because we use a certain precision level, does not guarantee that will be the level of precision we get as many factors can worsen a GPS location accuracy that are beyond the phone's control.

Android

Priority Level	Accuracy	Location Source	Battery Impacts
High Accuracy	Within 10m	Combined GPS, Wi-fi, cell tower	Highest energy consumption
Balanced	Within 100m	Combined Wi-fi & Cell towers	Low Energy Consumption
Low Power	Within 10km	Cell towers only	Very Low Energy Consumption
Passive	Unknown	Any available device	None

***Note** IOS works by setting a desired level of accuracy and then the device works to meet that target without exceeding it. Priority level names don't make any guarantee to meet the accuracy needed.*

IOS

Priority Level	Accuracy	Location Source	Battery Impacts
Best For Navigation	Highest Possible	Combined GPS, Wi-fi, cell towers & Apple "extra sensor data"	Tremendous. Recommended only when plugged into power source.
Best	Best Available without using "extra sensor data"	Combined GPS, Wi-fi, cell towers	High Demand
Nearest Ten Meters	Within 10m	Combined GPS, Wi-fi, cell towers	High Demand
Hundred Meters	Within 100m	Combined Wi-fi & cell towers, potential GPS use	Medium Demand
Kilometer	Within 1km	Combined Wi-fi & cell towers, potential GPS use	Medium Demand
Three Kilometer	Within 3km	Combined Wi-fi & cell towers, potential GPS use	Medium Demand
Reduced	Within 20km	None	None

***Note** The IOS "Reduced" level is a phone privacy setting that limits all your app's ability to detect your location. It has been included for completeness but is clearly not an option for us.*

What I didn't find:

There seem to be nearly 0 studies on the specific battery drain from using GPS. This may be due to the wide range of phones that a person could use as well as the highly unreliable nature of the quality of location data you are likely to get. Android advises testing a specific app with varying levels of GPS precision and basing decisions off that as opposed to calculating exact drain rates.

Recommendation:

I recommend using **High Accuracy on Android** and **Best on IOS**. The reason for our use of GPS is to identify where a near-miss happened and getting locations that are 100m off

could cause the wrong road to be identified or even hide available side-of-street information that a high precision location could give us. I see no meaningful impact from high precision requests that happen as infrequently as ours will occur. This could change as when we get GPS functional in app we can do practical tests and see if there is a difference in battery drain.

Low-Value Supportive Findings

I wrote an android app that makes high precision GPS calls four times a second. With that running it took 7 min and 51 seconds to lose 1% battery power. With GPS active, but when the app is not actively requesting precise data, a 1% drop in battery power took 8 minutes and 16 seconds. Most of the sources of drain were held to be consistent the entire time including screen intensity and background apps. While this is far from a formal benchmark, it does indicate that GPS calls have nearly no impact on battery drain. Our application will do 1 location call for each detected incident and potentially 1 every 30 seconds for approximating location after a disconnect. This represents a tiny fraction of the 1884 calls that I executed in 7 min and 51 seconds.

References:

All research was conducted on developer.android.com and developer.apple.com.